

2 TREES / THE BOOK

2 trees: Build your (D)Home

Working manuscript

Codex parallel draft

A reading edition of the current editable manuscript. Illustrations retain their captions and source notes.

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2 trees: Build your (D)Home

One small chainsaw. One hundred and twenty wedge members. Forty triangles. Two weeks to attempt a frame.

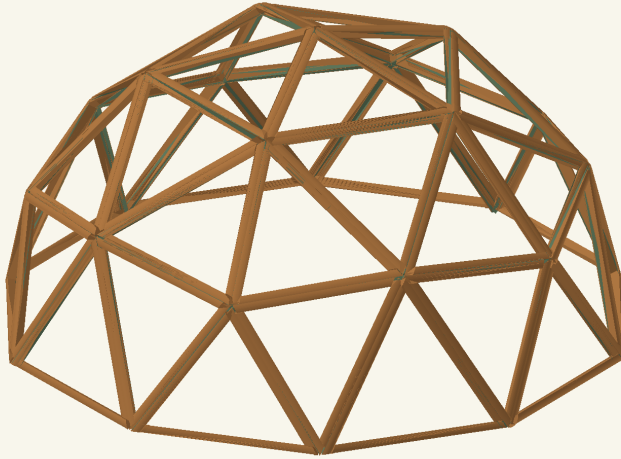
A practical explanation and a builder's field story, beginning with two trees and following the decisions that turn them into a geodesic dome.

This is the working manuscript. The construction described here is a planned build; its outcome will be recorded as it happens. The author describes the intended saw as a Husqvarna 120 Mark III purchased from Lowe's. The exact product label, receipt, and supplied bar-and-chain combination belong in the equipment record before publication.

2 TREES / PROJECT PLATES

Point Dome In

Forty independent pinwheel triangle frames; neighboring panels retain two members along each shared edge.



DomeSim geometry - illustrative model plate - dimensions and connection design remain separate

Figure 1

Raw-wedge dome: point toward dome interior; 120 independent panel members.

Source: Rendered from existing DomeSim model geometry; see `two_trees_codex/render.py`. Print colors are illustrative.

Before the first chapter

There are three ways to read this book. Follow the story to understand how the idea arrived. Follow the explanation to understand why the geometry works. Follow the how-to pages to organize measurements, prototypes, material, and assembly. Each chapter has a strand so that a reader can change pace without losing their place.

The title sets the scale of the experiment. Two trees are the intended source of the frame's wedge members. They do not include the ground connection, fasteners, seam components, skin, services, or everything required for a dwelling. The fortnight is a target for making and framing the dome, with its preparation stated openly. Occupancy is a later milestone with its own requirements.

The book uses **A for the long chord** and **B for the short chord**. Other references may reverse these letters. Compare lengths and factors before comparing names. Dimensions are inches unless marked feet; trunk diameter always means a measurement across the trunk, not the radius.

The number printed beside an ideal triangle is not automatically a saw cut. We will follow that distinction from the first sketch to the last stock check. A **blank** is the piece available to fabricate a member. A **member** is the finished wooden part. An **edge** is a line in the mathematical mesh. A **seam** is the interface between neighboring panels. Confusing these words is an easy way to build the right arithmetic into the wrong object.

The geometry in this draft traces to the project files listed in the source appendix. Practical prescriptions for structural connections must be supported by an appropriate design for the actual site and timber. The simulator establishes shape and fit; it does not establish allowable loads. This boundary is part of describing the project accurately.

PART / story / draft

Part I — The idea in the wood

Before a cut list, there was a question: what if the useful shape was already inside the tree?

We begin with the earlier experiments, then arrive at the wedge. This part explains the discovery without pretending that every later detail was obvious at the beginning.

1. Start with what is here

I want to find out how directly I can get from a tree to a frame. Not from a stack of uniform lumber, not from a kit delivered with every decision already made, but from the wood itself. The shape of that wood has to enter the design early enough to matter.

The proposal is small enough to count. Forty triangular frames need three members each: one hundred and twenty members. Two suitable trunks, each divided into eight useful sections, could yield eight wedge blanks per section. That is sixty-four blanks from each tree, sixty intended members and four potential spares. Those are the starting numbers, before I know which pieces the wood will actually permit.

The time limit gives the question an edge. I intend to design, produce the members, and frame the dome within two weeks, using the small Husqvarna saw described in my equipment notes. A limited tool makes the length and repeatability of the parts important. The saw's role belongs in the story because it affects the process; its model name is not a performance guarantee.

There is a temptation to tell the ending first. A photograph of a finished dome would make the early uncertainty disappear. I want to keep that uncertainty visible. Where does a nominal six-foot dimension become a real part? How much straight trunk is there after defects and cut losses? Which joint detail survives a full-size trial? Those questions are what the build is for.

The first record will be the starting inventory. Photograph both source trunks or the available sections. Include a tape in the frame. Record the actual usable lengths, the smallest diameters, and every assumption used to choose the initial dome. If the available wood changes the plan, the change becomes a chapter in the experiment.

2. What the Frankendome taught me

The earlier project notes describe the Frankendome as an experiment in joining unlike triangular frames. Its members were deliberately varied. The problem was not only how to make a triangle, but how one completed triangle could meet another when their edges did not offer identical profiles.

That experiment made the interface important. The notes describe testing V brackets at edge-to-edge and corner relationships. A connector that can accommodate an awkward member is useful because it lets the designer ask what the material can do before demanding that every piece become conventional rectangular lumber.

The notes also describe a modified upper pentagon: ten pieces in that earlier configuration where another arrangement would use fifteen. That number belongs to the earlier experiment. It must not quietly replace the forty-face, one-hundred-and-twenty-member count used for the book's standard panelized 2V hemisphere. A prototype can teach the next design without being the same design.

The useful inheritance is a way of thinking. Build manageable units. Give each unit a defined boundary. Study what happens at that boundary. If the connector and the member profile are designed together, the member has more possible shapes than the lumber aisle suggests.

For the final story, place the earliest dated Frankendome photograph beside a later wedge prototype. Describe one specific connection that was troublesome, what was tried, and what it revealed. The present draft has the sequence from the project notes; it still needs the author's remembered scene and the evidence that anchors it.

3. Finding the wedge

A round trunk contains eight equal radial sectors if its cross-section is divided at forty-five-degree intervals. Extend one sector along a straight section of trunk and it becomes a long member with two sawn radial faces and the original curved outside. The curve that rectangular milling would remove becomes part of the proposed frame.

The discovery in the project notes is the relationship between this profile and a panelized dome. A small triangular frame provides a repeatable use for the wedge. A separate seam detail can accommodate how neighboring triangles turn relative to one another. The member no longer has to do every geometric job by itself.

This is also where the attractive claims need to become questions. Preserving the curved portion can preserve material, but an area calculation alone does not say how much usable framing emerges from a real tree. The bark, defects, taper, irregular saw path, and the ends of the member still matter. The useful discovery is a method worth measuring, not a percentage protected from measurement.

In the design software, the wedge can face in four cardinal directions. Its point can face into the dome, out toward the sky, into its own triangular panel, or across the panel boundary. Each orientation changes the relationships at the seams and corners. The original outward-curve description is one configuration of the larger investigation.

The book's most useful discovery scene will show a change in understanding. Preserve a sketch or photograph of the first wedge beside the first assembled corner. Record what the wood suggested, what the model corrected, and what the full-size piece later changed again. Those three moments give the explanation a human sequence.

4. Define the experiment before starting the clock

Write the finish line in a sentence that someone else could check: produce and assemble the selected dome frame from documented source timber. Define which preparatory work happens before day one. Existing software development, drawings, fixtures, equipment purchases, site preparation, and professional design work may all be real contributions even when the cutting-and-framing clock has not started.

Use four milestone records: material available; accepted members produced; triangular panels checked; frame assembled and inspected against its design. A pile of ripped wedges reaches the first fabrication milestone. A standing frame reaches another. Neither should be counted as a finished home.

Keep an equipment ledger. The planned saw is identified by the author as a Husqvarna 120 Mark III from Lowe's. Record the nameplate and actual configuration rather than copying specifications from another generation of the 120. The manufacturer uses product-number information to distinguish variants; the purchase record resolves the particular machine in this story. Husqvarna product identification (<https://us-support.husqvarna.com/en/petrol-chainsaw/KA-70145>).

Also record the other work required to make the saw useful: measuring, securing and supporting work, moving timber, maintaining the tool, marking members, making fixtures, and managing the site. A small saw does not make those operations disappear. It makes the organization around them especially visible.

Before field work, use training appropriate to the operations and the manufacturer's instructions for the exact equipment. Protective equipment, secure work support, separation from helpers, and emergency

arrangements belong in the work plan. Felling and handling timber under load require skills that a chapter about dome geometry cannot supply. The Forest Service training material treats saw use as both classroom and field instruction. USFS saw training guide (<https://www.fs.usda.gov/t-d/pubs/htmlpubs/html06672805/page01.htm>).

PART / explanation / draft

Part II — The dome before the saw

A sphere can be imagined continuously. Wood must arrive as countable pieces.

This part takes the project from a radius to a mesh, then from a mesh to independent wooden panels. Keep the reference geometry visible while the physical object becomes more complicated.

5. A sphere made from straight lines

A geodesic construction begins with a regular organization of directions on a sphere. In this project's 2V model, the triangular faces of an icosahedron are subdivided once and the resulting points are projected to the sphere. The chords between those projected points become the straight edges of the triangulation.

The frequency describes subdivision. It does not say how many feet across the dome is. A 2V dome can be enlarged or reduced while retaining its geometric relationships. Every nominal edge scales with the sphere radius, which is why a dimensionless chord factor can connect a desired dome size to a length on the bench.

The hemisphere used here contains forty triangular faces, twenty-six vertices, and sixty-five unique edges. Its open rim is a ten-sided polygon. The vertices on that rim lie on a circle; the floor boundary between them is made from straight lines. This difference matters when quoting floor area.

The model separates two edge classes. With the book's lettering, A is the longer chord and B is the shorter. Ten panels have three A sides. Thirty panels have one A side and two B sides. The corresponding triangle shapes repeat, but the real wooden members also depend on how their corners and seams are made.

Look at the topology plate before the timber plate. In the topology, a line has no thickness, no end cut, and no fastener. Its value is precision: it establishes exactly what the assembly is meant to follow. The wooden model adds the decisions necessary to occupy space around those lines.

6. Why 65 edges need 120 pieces of wood

The dome's shared-edge mesh and its independent-panel frame count different things. If each mathematical edge owned one stick, the hemisphere would use sixty-five sticks. In this project, each of forty triangles owns all three of its wooden members. The frame therefore contains $40 \times 3 = 120$ members.

Panel family	Panels	A members per panel	B members per panel	A total	B total
A-A-A	10	3	0	30	0
B-A-B	30	1	2	30	60
Total	40	—	—	60	60

Two neighboring triangles bring two pieces of wood to their common boundary. The duplication is intentional. Each triangular unit can be made and checked before it joins the dome. The space between its member and the neighboring panel's member becomes the seam detail.

Of the sixty-five unique mesh edges, ten are at the exposed base rim and fifty-five lie between two panels. Count forty panels $\times$ three edges to get one hundred and twenty panel-edge incidences. Count the same incidences another way: fifty-five interior edges appear twice and ten boundary edges appear once. Thus $55 \times 2 + 10 = 120$.

There are thirty-five long edges and thirty short edges in the unique-edge mesh. Ten of the long edges are on the base. The shared seams are therefore twenty-five long and thirty short. Use the member count to plan wedge stock; use the seam count to plan panel interfaces. Hardware quantities require the selected detail at each interface and cannot be inferred by calling all edges “struts.”

This page should remain beside the cut schedule. When another calculator returns thirty-five long and thirty short, it may be correctly counting the conventional shared-edge dome. The disagreement can be a difference in construction system rather than an arithmetic mistake.

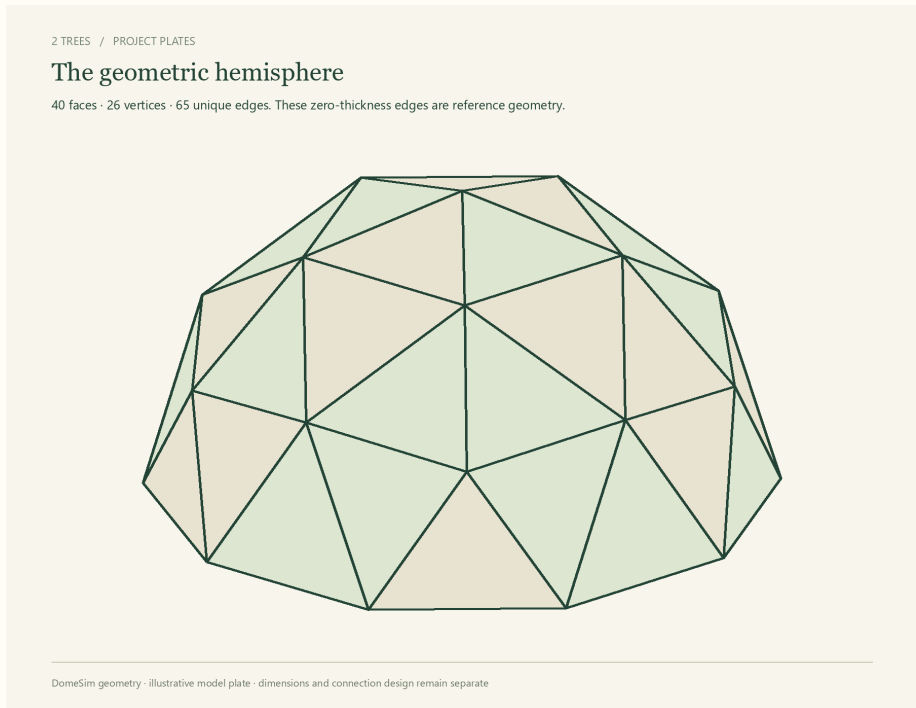


Figure 1

2V reference topology: 40 faces, 26 vertices, 65 unique edges.

Source: Rendered from existing DomeSim model geometry; see `two_trees_codex/render.py`. Print colors are illustrative.

7. Method one: start with the radius you want

Choose a sphere radius R , then keep one unit throughout the calculation. If the radius is in feet and the working dimensions will be inches, multiply it by twelve before finding the chords.

The nominal relationships are $A = R \times f_A$ and $B = R \times f_B$. For this project's exact generated 2V mesh, f_A is approximately 0.618033989 and f_B is approximately 0.546533058. The Sizing lab obtains these factors from the existing geometry module rather than keeping the rounded printed values as its authority.

The procedure is sequential. Record the desired radius. Generate both chord lengths. Confirm the forty-panel inventory. Select the member cross-section, orientation, and seam construction. Generate the physical model and inspect its actual stock lengths. Only then select raw blank lengths and allocate trees.

If a radius gives nominal edges that are inconvenient for the timber, the design still has options. A slightly smaller radius can reduce length demand. A different seam or corner setting can change the stock envelope. More usable source length may be available in a longer section. Each option must be checked through the same solver; none should be hidden inside an unexplained rounding choice.

Record three lengths for each member: the ideal chord, the finished physical stock envelope, and the blank required to fabricate it. The last includes whatever additional holding, trimming, and overfit stock the actual process needs. Zero extra allowance in the lab explicitly means that only the finished envelope is being checked.

When the stock plan is complete, export the actual simulator's bill of materials and cut schedule for the chosen configuration. The book can

explain the reasoning, while that schedule preserves the geometry used for the build. Write the configuration and date beside any printed schedule so an older paper does not silently govern a newer design.

8. Worked example: a ten-foot radius

Use a sphere radius of 10 feet, or 120 inches. The nominal diameter is 20 feet. Multiply the exact mesh factors by 120 to obtain a long chord of approximately 74.164 inches and a short chord of approximately 65.584 inches.

The panelized frame requires sixty members associated with A edges and sixty associated with B edges. These labels identify edge families, not necessarily just two physical cut lengths. A member's surrounding triangle and joint relationships can split a nominal family into multiple fabrication cases.

The enclosing base circle has area $\pi \times 10^2$, approximately 314.159 square feet. The straight-sided decagonal base has area $5 \times 10^2 \times \sin(36^\circ)$, approximately 293.893 square feet. Neither quantity is the finished usable floor area after wall build-up, foundations, doors, and fittings. The circle is a useful reference; the polygon describes this mesh's base.

For a concrete physical comparison, the current raw-wedge solver with an eight-inch trunk diameter, point facing into the dome, raw-trapezoid seams, and no additional fabrication allowance produces a longest finished stock envelope of approximately 72.952 inches at this radius. That is different from the 74.164-inch nominal A chord because the member's actual ends and offsets are solved separately.

This example reveals why a six-foot blank cannot simply be assigned to a “six-foot dome edge,” and why the error does not always have the same sign. Some physical parts are shorter than the nominal edge; different settings and additional holding stock can change what the blank must provide. Inspect the physical schedule instead of assuming a fixed deduction.

The Sizing lab reproduces this example from the repository's solver. If the solver is revised, insert a fresh result page and retain the former page as a dated record. The published example should match the configuration and code version used to fabricate the actual frame.

9. Method two: let the tree set the size

Begin with a measured inventory rather than a desired radius. Identify usable straight sections in the available trunks. Record their lengths, their smaller-end diameters, and the defects that affect where a member can be obtained. A tall tree and a long run of usable stock are different measurements.

The book's starting proposal is eight sections of approximately six feet from each tree. Eight radial blanks per section produce $8 \times 8 = 64$ potential blanks from one tree. Reserving four for rejection or spares leaves sixty intended members. Two trees produce 128 potential blanks and a target allocation of 120 members.

This is a count-based first pass. A blank earns its place in the frame only after it meets the length, cross-section, straightness, and material requirements for its assigned part. Eight adequate pieces at the butt of the tree do not prove that eight adequate pieces also exist in every upper section.

Let S be the usable blank length after planned end trimming. A nominal-only starting estimate is $R = S / fA$. If S is 72 inches, that gives a radius of approximately 116.498 inches, or 9.708 feet, and a diameter of approximately 19.416 feet. The corresponding nominal short chord is about 63.670 inches.

That estimate intentionally ignores physical corner and seam geometry. To size the real frame, hold the selected cross-section and joint settings fixed and solve for the largest radius whose every required blank fits the available length. The Sizing lab's "Fit physical stock" action uses the raw-wedge solver for this check. Enter any additional fabrication stock

rather than treating the final installed piece as the whole blank requirement.

The two methods meet here. Design-first asks how much wood a chosen radius requires. Tree-first asks what radius the available wood permits. With identical joint settings and allowances, going forward and then backward should return the same design. That round trip is an unusually useful check on a cut list.

10. Eight sections, sixty-four blanks, sixty accepted members

Quantity	One tree	Two trees
Planned six-foot sections	8	16
Potential blanks at eight per section	64	128
Target allocation	60	120
Potential spare or rejected blanks	4	8

The eight potential spares represent 6.25 percent of the gross 128-piece inventory. If eight blanks are rejected, the remaining 120 exactly meet the count and leave no accepted replacements. At a ten-percent rejection rate, the arithmetic no longer supports a complete 120-member frame from this inventory. Report that shortage while there is still time to change the plan.

Eight six-foot sections contain forty-eight feet of net section length. They cannot all be cut from exactly forty-eight feet while also losing material to separating cuts. If seven interior bucking cuts remove a quarter inch each, the minimum span before outer-end trimming is 48 feet plus 1.75 inches, or about 48.146 feet. Additional cuts at the ends and defect removals require more length. This example declares its kerf assumption; it is not a measurement of the author's saw.

Separate bucking kerf from ripping kerf. The first reduces the longitudinal inventory as sections are separated. The second removes wood from their cross-sections as wedges are produced. Both matter, but subtracting one from the other would mix different dimensions.

One organizational option is to reserve one tree's qualified blanks for A members and the other tree's for B members. That makes the title easy to explain, but assignment should follow the actual inventory. A short or defective A candidate may be perfectly useful in a B position if its material properties and finished dimensions meet the design. Keep its original tree and section ID when reassigning it.

The project's earlier timber demonstration also explores twelve-foot sections split and then divided into shorter members. That is a different cut plan. Its yield cannot be copied onto this eight-by-six-foot plan without accounting for usable trunk length and every additional crosscut.



Figure 1

Eight sections per tree, eight potential blanks per section: 128 gross blanks and a target of 120 members.

Source: Rendered from existing DomeSim model geometry; see `two_trees_codex/render.py`. Print colors are illustrative.



Figure 2

Eight six-foot sections, eight potential wedges per section: 64 gross blanks per tree and 128 from two trees. The 120-member frame target leaves eight gross blanks before real defects and losses are counted.

Source: Rendered from Two Trees, All at Once / explode at 88%, using the original video scene and camera (original overlay). Source model assumptions, not verified field results. Timing uses authored chapter durations; source hashes and the complete snapshot recipe are saved with this image.

PART / explanation / draft

Part III — The tree becomes inventory

The model contains ideal circles. The source timber contains taper, history, and surprises.

This part turns the attractive gross count into a traceable inventory of usable pieces. The small section is valuable because each decision can be local, visible, and recorded.

11. The shape of one eighth

For a circular trunk of radius r , one ideal eighth-sector has an angle of 45 degrees and area $\pi r^2/8$. Its two radial sides each have length r . The outside is an arc, not a straight third side. Its arc length is $\pi r/4$; the straight chord joining the arc's endpoints is $2r \sin(22.5^\circ)$.

For an eight-inch diameter trunk, r is four inches. The ideal sector area is approximately 6.283 square inches. The curved arc is approximately 3.142 inches long, while its straight chord is approximately 3.061 inches. These describe a raw circular sector before kerf, bark removal, or other changes. Drawing a triangle in its place removes some of the wood the idea is trying to preserve.

For a twelve-inch diameter trunk, the ideal sector area grows to approximately 14.137 square inches. A six-foot length of that ideal sector contains about 1,017.9 cubic inches, or 7.07 solid-volume board feet. Board feet here are only a volume unit: 144 cubic inches per board foot. This calculation is not a lumber grade or a market valuation.

A wedge can have substantial area while still presenting difficult bearing surfaces and connections. Its orientation changes how it occupies the panel boundary. Its taper changes the section along its length. Its grain and defects change the material that carries load. The book therefore uses the cross-section to explain the shape and the inventory, while keeping strength evaluation attached to the actual member and design.

The illustration on this page should show a whole log cross-section beside one true sector, then the same sector oriented in the four ways explored by the simulator. Use end-grain views. A long perspective drawing can hide the very profile the reader needs to understand.

12. Read the trunk in sections

Give each source tree an ID before allocating members. Give each candidate section a second ID and mark its original orientation. Then record usable length, large-end diameter, small-end diameter, visible sweep, knots, checks, damage, and any reason a region must be excluded. Photograph the section beside its label.

A short-section method can respond to local curvature. If a bend interrupts a long run, the useful material on either side may still supply shorter members. This is a possibility to inspect, not a guarantee that any crooked trunk can be converted into the planned stock count. Every new boundary also consumes length through a cut and may create an end that needs trimming.

Use the smallest relevant cross-section when checking whether a blank provides the material the design requires throughout its length. An impressive butt diameter is not the diameter of an upper six-foot section. A circular model also needs to be reconciled with an oval or irregular real section before its area becomes an inventory claim.

Make assignments provisionally. A long, sound blank can remain uncommitted until the most demanding members are covered. Shorter acceptable blanks can be considered for the B family. Pieces awaiting judgment should stay in a separate recorded category so that uncertain stock does not appear in the accepted count.

The field ledger is more useful than a single yield percentage. Its rows can answer where a missing member went. Was it lost to end damage, excessive taper, a defect exposed during cutting, or a fabrication error? Those explanations turn an experiment into a process another person can evaluate.



Figure 1

The starting inventory is a standing pine. A still from the existing harvest lesson, captured at 30 percent of its opening tree scene; the on-screen number illustrates the source lesson, not a measured harvest result.

Source: Rendered from Two Trees, All at Once using the existing scene painter and authored camera. Source-model example.

13. Measure recovery instead of defending a percentage

The discovery notes propose a high wood recovery and compare it with conventional milling. That is a useful hypothesis. To test it, define the numerator and denominator before looking at the result. Gross solid trunk volume, sawn sector volume, accepted member volume, and installed member volume are different quantities.

For a simplified circular section, the total kerf area of four diametral cut planes can be estimated by $4Dk$, where D is diameter and k is kerf width. Compared with the round area $\pi D^2/4$, the simple retained fraction is approximately $1 - 16k/(\pi D)$. The approximation ignores overlaps between cut bands near the center and assumes the intended cut geometry. Actual kerf paths, bark, irregular shape, end losses, and rejected wood change the result.

At $D = 12$ inches and $k = 0.25$ inch, this simple estimate retains approximately 89.39 percent before those other losses. At $D = 8$ inches with the same kerf assumption, it retains approximately 84.08 percent. The change with diameter is enough to show why one attractive percentage should not stand in for an entire tapered tree.

The earlier project timber model computes a tapered trunk by sections and compares its wedge output with its own rectangular packing model. Record that comparison's assumptions: section length, kerf, taper, nominal or actual rectangular dimensions, and acceptance criteria. A result for true two-by-fours is not automatically a result for dressed nominal two-by-fours.

The book's final comparison should use the same starting inventory on both sides. Include labor, equipment, defects, and the material that genuinely replaces a required product. A wedge with more volume than a rectangular stick is not automatically worth that volume multiplied by a store price. It must perform the intended job through an appropriate design.

The author can make the strongest account by keeping the unfriendly rows: pieces rejected, setup that took longer, and usable stock left over. Those rows show where the method can improve and prevent a gross-material argument from pretending to be a finished-building argument.

14. Wood keeps changing after the cut

The model's geometry stays still after a calculation. Freshly cut timber continues to exchange moisture with its surroundings. Changes in dimensions, shape, and checking can affect joints that fitted at the first assembly. The field record therefore needs dates and moisture observations, not just a photograph of a tight new corner.

Record the timber species when it is reliably identified, the condition of the stock, how it was stored, and any moisture readings with the method used. Preserve examples of the cross-section at fabrication and at later inspection. When a seam changes, this record helps distinguish an initial cutting error from movement after assembly.

The USDA Wood Handbook organizes moisture relations, mechanical properties, fastenings, grading, and drying as separate but related subjects. Those are the subjects the prototype's material record needs to connect.

USDA Wood Handbook

(<https://research.fs.usda.gov/fpl/wood-handbook>).

Avoid turning an attractive green-wood fit into a prediction of long-term performance. The final fastening and seam design needs to account for the actual timber, exposure, and changes over time. The two-week schedule does not erase those later observations; it gives them a clear initial condition.

Reserve a few documented samples for follow-up. Mark them with the same tree and section IDs used in the member ledger. If a sample changes while the frame is in service, it becomes a small, accessible witness to the larger material process. Its behavior is evidence to interpret, not a substitute for inspecting the actual frame.

PART / how-to / draft

Part IV — One hundred and twenty useful pieces

Repeatability is the bridge between a promising shape and a buildable set of panels.

This part moves from a representative blank to a checked corner, then to a complete triangle and a controlled batch. Each stage earns the right to repeat the previous one.

15. Organize the work around the small saw

The intended tool is part of the constraint that makes the experiment interesting. It is also a machine with a defined operating envelope. Record its actual model, approved equipment combination, condition, and maintenance. Use its manual and appropriate instruction to determine operations and setup; do not infer capability from an illustration or from bar length alone.

The planning sequence begins with stock that has been safely obtained, supported, and made accessible. Divide the work into measurement and marking, material processing, inspection, and storage. Give each activity enough room that an interruption does not put an unmarked or uninspected member into the accepted stack.

Before committing the inventory, document one representative sample through the proposed process. Measure the resulting kerf and the usable piece. Record the time spent securing and repositioning work, maintaining the chain, and clearing material. Cutting time alone describes only one part of the operation.

Keep the production log short enough to use. A row needs the source section, resulting blank IDs, accepted count, rejected count, elapsed labor, and reason for losses. A photograph of the labeled group supports the row. If a process change improves the result, label when it started so the book can compare like with like.

The goal is a controlled batch, not a dramatic image of hurried cutting. The story can include fatigue, a dull chain, weather, or a setup that had to be changed if those events occur. They belong in the account because they explain the labor needed to transform a counted tree into counted parts.



Figure 1

A snapshot of the existing felling animation. The scene depicts the change from standing tree to felled trunk; it is a visual explanation from the source film.

Source: Rendered from Two Trees, All at Once / fell at 62%, using the original video scene and camera (original overlay). Source model assumptions, not verified field results. Timing uses authored chapter durations; source hashes and the complete snapshot recipe are saved with this image.

16. Four ways to place the same wedge

The raw-wedge simulator names each orientation by the direction of the sector point. **Point dome in** directs the point toward the sphere center. **Point dome out** turns it toward the exterior. **Point panel in** turns it toward the center of its own triangular panel. **Point panel out** directs it across the panel boundary.

These labels describe cross-section orientation around the member's long axis. They do not rename the geodesic edge. An A edge remains an A edge while the wood surrounding it changes position. The change can alter which face is available for a seam or a neighboring member's end.

Compare all four with identical nominal dome size, trunk diameter, and seam settings. Use a wide view to see the overall frame and a close end-grain view to understand the interface. Record the configuration on the figure rather than relying on a nickname such as “inside-out.”

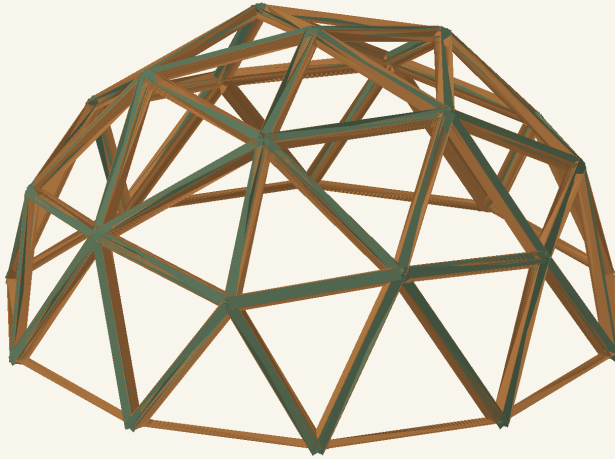
The project's original specification uses the curved surface toward the sky and the point inward. Later tools explore the other orientations. The narrative should make this progression visible: the first idea established a starting configuration; inspection then opened a larger design space.

Selecting an orientation is a fabrication decision. Once a representative full-size panel has been checked, mark that orientation on each blank and in the fixture documentation. A correct length installed with an unintended quarter-turn is a different physical assembly.

2 TREES / PROJECT PLATES

Point Dome Out

Forty independent pinwheel triangle frames; neighboring panels retain two members along each shared edge.



DomeSim geometry - illustrative model plate - dimensions and connection design remain separate

Figure 1

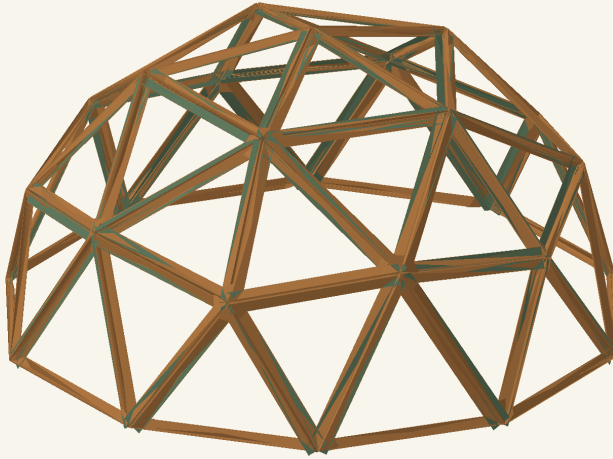
Raw-wedge dome: point toward exterior.

Source: Rendered from existing DomeSim model geometry; see `two_trees_codex/render.py`. Print colors are illustrative.

2 TREES / PROJECT PLATES

Point Panel In

Forty independent pinwheel triangle frames; neighboring panels retain two members along each shared edge.



DomeSim geometry - illustrative model plate - dimensions and connection design remain separate

Figure 2

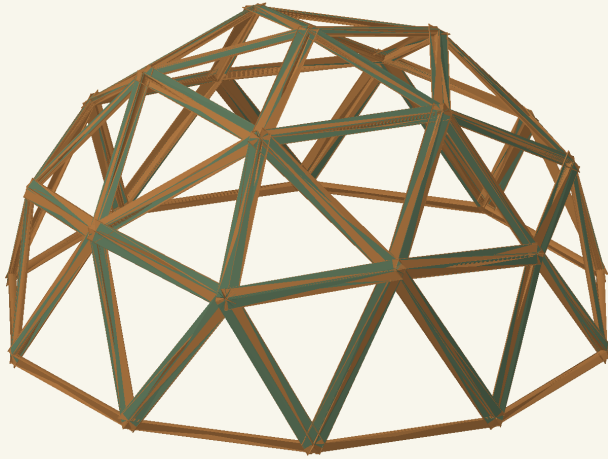
Raw-wedge dome: point toward the triangular panel interior.

Source: Rendered from existing DomeSim model geometry; see `two_trees_codex/render.py`. Print colors are illustrative.

2 TREES / PROJECT PLATES

Point Panel Out

Forty independent pinwheel triangle frames; neighboring panels retain two members along each shared edge.



DomeSim geometry - illustrative model plate - dimensions and connection design remain separate

Figure 3

Raw-wedge dome: point across the panel boundary.

Source: Rendered from existing DomeSim model geometry; see `two_trees_codex/render.py`. Print colors are illustrative.

17. The triangle corner is a physical object

The ideal triangle has vertices where lines meet. Real members have width, depth, and ends that must occupy different space. The project's pinwheel arrangement gives each member a receiving role at one end and a terminating role at the other. One member ends against the side of its neighbor while receiving another member in turn.

Earlier explanatory code describes simplified square-ended pinwheel members. The current raw-wedge simulator carries more detailed corner geometry: the terminating end is solved against its receiving surface, and the other end is clipped to the triangle's envelope. The corresponding cut planes and stock extents should govern the fabrication discussion for this configuration.

This correction matters to the story. Preserving the longitudinal forty-five-degree sector does not imply that every end is a square crosscut. The long radial faces, the panel-to-panel fold, and the member's end cuts are separate geometric questions. A process can preserve the raw sector while still requiring an oblique end cut.

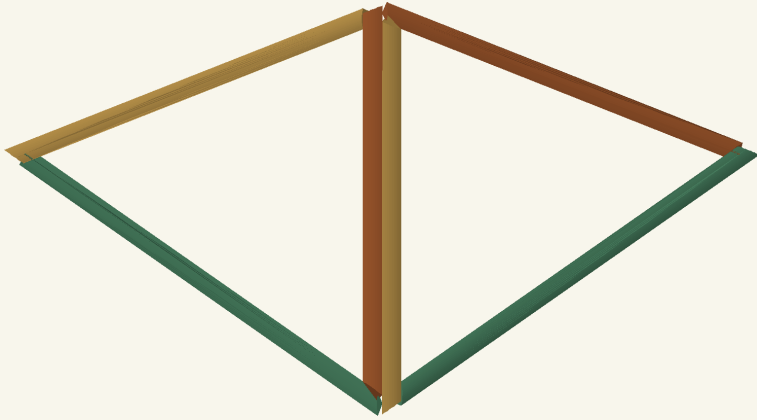
Start with a full-size representative corner using the intended cross-section and orientation. Mark the mathematical reference points and the actual cut surfaces separately. Check the receiving relationship, the available bearing region, the fastening space, and whether either member extends beyond the intended panel envelope.

Move from that corner to one A-A-A panel and one B-A-B panel. Check their reference dimensions and their physical boundaries. A fixture that reproduces one family does not automatically reproduce the other. Before cutting the batch, retain a labeled master example and the configuration that produced it.

2 TREES / PROJECT PLATES

The pinwheel / two neighboring panels

Each butt meets the side of the next member. Each triangle owns three members; a shared axis carries two.



DomeSim geometry - illustrative model plate - dimensions and connection design remain separate

Figure 1

Two separate cyclic pinwheel triangles, each with three members. Two members remain at their shared axis. Colors identify member ownership.

Source: Rendered from existing DomeSim model geometry; see `two_trees_codex/render.py`. Print colors are illustrative.

Plate — Two triangles, two members at the seam

Follow the three differently colored members around either triangle. Each member receives its neighbor near one end and terminates against the side of the next. The two central members are separate pieces, one owned by each panel. This is the cyclic pinwheel arrangement clarified by the author.

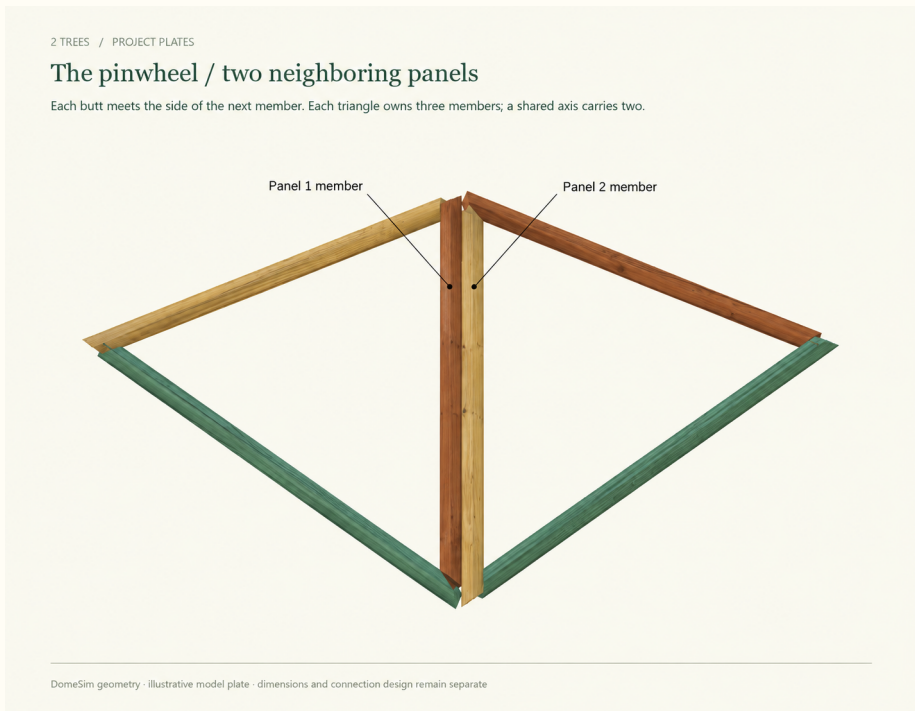


Figure 1

Each central member belongs to a different triangle. Around each triangle, the butt of one member meets the side near the front of the next.

Source: AI-finished illustration using the actual simulator pinwheel-pair render as the geometry reference; original model plate retained in Chapter 17.

18. Build the reference into the fixture

A fixture is a way to make the important measurements repeatable. Begin with the selected triangle's reference geometry, then establish stops, registration surfaces, and cut references from the actual physical model. The fixture should make it clear which part of the irregular wedge is being located and which remains free to vary.

The simulator's jig stages are useful illustrations because they separate setup, placement, and trimming. Review them with the chosen model. Do not transfer a picture's apparent dimension onto wood: use the current cut schedule and a verified full-size layout.

Write down the allowance used at each end. The model includes concepts such as head overfit and butt hangoff. Those are deliberately sacrificial pieces of stock, and their values can make a blank longer than the final member. A six-foot finished part with additional holding stock requires more than six feet of raw material.

After the representative panels are assembled, compare them with the reference layout. Check edge references, corner positions, plane of the frame, and the surfaces that will meet adjacent panels. Record the acceptance criteria from the actual design. “Looks triangular” is not a measurement a second builder can reproduce.

The fixture record should contain an overhead photograph, the version of the layout, the panel family, the wedge orientation, the stock cross-section used for the trial, and any adjustments made after the first piece. If a stop moves during the batch, identify which parts were made before and after that change.

The fixture is also a storytelling object. It shows the moment the project becomes a repeatable process. A reader can see how the useful idea was

transferred from screen to bench, and which details had to change when the material became real.

19. Make forty triangles, one checked batch at a time

Build the inventory around two panel families: ten A-A-A frames and thirty B-A-B frames. Preserve the member IDs through assembly so each completed panel can be traced to its source sections. A panel label should identify its family, handedness, orientation, and position in the assembly plan.

Treat the first checked panel of each family as a reference, not as permission to stop measuring. Recheck the fixture after adjustments and at sensible batch intervals. When a piece is rejected, keep the reason in the log. Reworking it, replacing it, and accepting it unchanged are different outcomes.

The short family is not simply whatever remains after making long members. Each B member still needs the right usable section and joint geometry. Allocate the demanding pieces first, then review how the remaining inventory can satisfy the rest of the schedule. The source tree assignment can change while the member's required geometry stays fixed.

At the end of each batch, update three counts: blanks remaining; finished accepted members; completed accepted panels. This avoids the common error of counting the same wood once as a member and again as part of a panel. Keep rejected and experimental material outside those totals.

Photograph completed panels flat from overhead and with a low angle that reveals their profile. Include a scale reference and a readable label. Those images serve the book's explanatory pages, but their first job is to document what was actually built.

Before transport or raising, check the completed inventory against the assembly map and the design's handling plan. Panels have mass and can be awkward to control even when each member was manageable alone. The

modular method changes the unit being handled; the handling arrangements must change with it.

PART / how-to / draft

Part V — The seams make the dome

Forty correct triangles are an inventory. Their connected relationships make the frame.

The next chapters follow those relationships from the shared seam to the ground and then to the larger question implied by the word home.

20. What happens between two panels

At an interior edge, two outward panel normals point in slightly different directions. The small angle between them is the fold: $F = \arccos(N_1 \cdot N_2)$ for normalized normals, with the numerical dot product clamped to the valid range. The internal supplementary angle is 180 degrees minus F . Print the definition beside the number because “dihedral” is used in different ways.

The raw sector contributes its own fixed face relationships. In the original point-inward comparison, the ideal residual gap angle is related to 45 degrees minus the panel fold. The actual contact surfaces, offsets, and selected seam mode must still be solved in their physical positions. An attractive sketch does not define a fabrication polygon.

The project's seam studies include rigid keys, hose or gasket concepts, and a shaved-flat alternative. The current raw-trapezoid mode preserves the longitudinal sector faces and uses a truncated key profile. The shaved-flat configuration changes those seam faces and uses a different key. These are distinct construction choices and should have separate captions and schedules.

A rigid shape can help establish a geometric relationship when the surfaces meet as intended. A round hose or gasket mainly fills space and can permit relative rotation unless the connection independently constrains it. Filling a gap, setting an angle, transferring structural forces, and managing water are different jobs. A single interface may need several components to do them.

Make a representative two-panel seam before the larger assembly. Use the actual timber profiles and the proposed connection detail. Record the fit, the available fastening geometry, and the relationship to the reference fold. This trial helps locate a problem while it can still be handled on the bench.

21. Give the frame a place to stand

The base ring is a ten-sided boundary in the selected hemisphere. It must be laid out from a consistent center and reference dimensions. A radius drawn on the ground locates the circumcircle; the segments joining the base vertices locate the polygon. Record which reference was used at every check.

The structural path continues from the panels through their connections and into the support and anchorage system. The base is therefore part of the design, even though the two trees are supplying the wedge members above it. Record the actual foundation or support approach, site information, and the design responsible for uplift, sliding, settlement, and the applicable loads.

Check the base layout before accepting panel fit as evidence about the dome. A misplaced support can make correct panels appear wrong. An improvised adjustment in one seam can then distribute that error through several others. Keep the layout measurements separate from the panel acceptance measurements so the source of a mismatch can be found.

The book should include a plan-view plate with base vertex IDs and reference dimensions from the chosen model. Pair it with an actual site photograph showing the same labels. This pair is more useful than a generic rendering because it connects the drawing to the place where the assembly must close.

Any doorway that interrupts the frame or base changes the local arrangement. Describe the intended opening as an explicit design variant. Do not silently remove members from the forty-triangle scheme while continuing to quote its original topology and stock count.

22. Follow an assembly plan that can stay stable

The assembly order must come from a plan for the actual panels, connections, equipment, and site. A complete dome's behavior does not describe an incomplete ring of panels. Temporary support, controlled handling, and the limits of each intermediate stage need their own attention.

Before raising begins, lay out the panel map and verify every label against it. Identify each planned seam, the connection components it requires, and the stage at which it can be checked. Record where temporary support is required and what inspection permits a stage to proceed. These details should be supplied by the build's actual assembly and structural plan.

The earlier Frankendome notes discuss a top-down experiment. That history belongs in the narrative, but it does not choose the safe order for this different frame. The present book should show the sequence actually adopted, with its equipment and supports visible in the photographs.

As the frame grows, compare reference nodes and the base with the model. A forced seam is useful information. Pause to identify whether the cause is the base layout, a panel dimension, an orientation error, or a connection mismatch. Record the diagnosis and correction so the story preserves the discovery rather than editing out the difficulty.

The closure is an important scene only if its conditions are visible. Photograph the final relationship before and after connection, record any adjustment, and retain the final dimensional checks. The reader should be able to understand what “it fit” meant beyond the relief of seeing the last piece in place.

23. The frame and the home it might become

The parenthesis in (D)Home keeps two scales in view. This project is making a dome frame. The larger ambition is useful shelter shaped by that frame. The next decisions include the envelope, openings, drainage, insulation, ventilation, services, interior space, and maintenance access.

The project already explores many of these subjects. Dome Creator can show frames, panels, interiors, and site arrangements. Other lessons examine energy, material quantities, assembly flow, and alternative shapes. Use those tools to illustrate questions the frame creates, while labeling conceptual configurations so a reader does not mistake an image for a completed building detail.

The seam has to be considered again when a skin is introduced. Water follows exterior geometry and discontinuities, not the simplicity of a frame diagram. An opening introduces edges where layers end or turn. A service penetration has to be coordinated with the envelope and with the structure. Those are design topics to develop explicitly in a later phase.

Space also deserves a measured description. The base polygon, the curved wall profile, and the location of the useful head-height zone influence furniture and circulation. A floor area quoted from an enclosing circle tells only part of that story. Include a plan and a section when describing a proposed interior.

At the fortnight's end, name the achieved milestone precisely. If the frame is standing, describe that accomplishment and its remaining work. If a problem has delayed the frame, describe the problem and what the experiment established. The title can hold ambition without requiring the field record to invent completion.

PART / story / draft

Part VI — Fourteen days, as they happen

The calendar is a promise to keep a record.

The following pages begin as a proposed schedule. Replace the placeholders with actual dates and observations. Move work when the project requires it and preserve the reason for the move.

24. Plan the fortnight, then record the difference

The schedule is a working hypothesis. It assumes the source timber, site arrangements, appropriate design decisions, equipment, and relevant preparation are available. List preparation honestly beside day one. Work that took place before the clock still belongs to the cost and history of the project.

Days	Proposed focus	Evidence that closes the stage
1–2	Verify source inventory and freeze the first physical configuration	Measured section ledger and saved model
3–4	Trial stock processing and representative corners	Accepted samples, measured losses, revised allowance
5–7	Produce, inspect, and assign the remaining wedge blanks	Actual accepted member inventory
8–10	Build and check both triangle families	Panel register and inspection record
11	Verify the base, equipment, connections, and assembly readiness	Stage review against the actual assembly plan
12–13	Assemble through planned supported stages	Stage photographs, corrections, dimensional checks
14	Record the achieved condition and the remaining work	Final inventory, photographs, and truthful outcome

These are allocations, not daily quotas that override conditions. An unresolved trial corner is a reason to move the batch, because repeating an uncertain detail produces more uncertainty. An interruption should appear in the schedule rather than be hidden by counting only the hours when the saw was running.

Each field page asks for the same small set of facts and one different story question. The facts make the process comparable. The question keeps the narrative from becoming only a spreadsheet. Include a photograph that proves a change, even when the change is a correction or a rejected piece.

Day 01 — The starting inventory

Planned focus

Measure and label the available sections; save the first configuration.

This is a proposed focus, not a claim that the work occurred on this day.
Record the actual sequence below.

Field facts

Date: [record]

Start / finish / breaks: [record]

People and labor hours: [record]

Weather and site conditions: [record]

Starting inventory -> accepted output -> rejected / reworked -> remaining:
[record]

Measurements and configuration used: [record]

The scene

What did the actual trees change about the initial plan?

[Write the actual scene in your own words: what you saw, tried, changed, and understood.]

Evidence and next step

Photo or video IDs, captions, receipts, and notes: [record]

What changed in the plan and why: [record]

Next step: [record]

Day 02 — The first configuration

Planned focus

Compare desired-radius and available-stock routes; record allowances.

This is a proposed focus, not a claim that the work occurred on this day.
Record the actual sequence below.

Field facts

Date: [record]

Start / finish / breaks: [record]

People and labor hours: [record]

Weather and site conditions: [record]

Starting inventory -> accepted output -> rejected / reworked -> remaining:
[record]

Measurements and configuration used: [record]

The scene

Which dimension became the controlling constraint?

[Write the actual scene in your own words: what you saw, tried, changed, and understood.]

Evidence and next step

Photo or video IDs, captions, receipts, and notes: [record]

What changed in the plan and why: [record]

Next step: [record]

Day 03 — The first useful blank

Planned focus

Record the representative stock trial, actual losses, and setup time.

This is a proposed focus, not a claim that the work occurred on this day.

Record the actual sequence below.

Field facts

Date: [record]

Start / finish / breaks: [record]

People and labor hours: [record]

Weather and site conditions: [record]

Starting inventory -> accepted output -> rejected / reworked -> remaining:
[record]

Measurements and configuration used: [record]

The scene

What was different between the imagined wedge and the real one?

[Write the actual scene in your own words: what you saw, tried, changed, and understood.]

Evidence and next step

Photo or video IDs, captions, receipts, and notes: [record]

What changed in the plan and why: [record]

Next step: [record]

Day 04 — The first corner

Planned focus

Document end-to-side contact and the cyclic pinwheel with close photographs.

This is a proposed focus, not a claim that the work occurred on this day.
Record the actual sequence below.

Field facts

Date: [record]

Start / finish / breaks: [record]

People and labor hours: [record]

Weather and site conditions: [record]

Starting inventory -> accepted output -> rejected / reworked -> remaining:
[record]

Measurements and configuration used: [record]

The scene

What did the contact between two members teach you?

[Write the actual scene in your own words: what you saw, tried, changed, and understood.]

Evidence and next step

Photo or video IDs, captions, receipts, and notes: [record]

What changed in the plan and why: [record]

Next step: [record]

Day 05 — A repeatable process

Planned focus

Record fixture changes and separate accepted, uncertain, and rejected pieces.

This is a proposed focus, not a claim that the work occurred on this day.

Record the actual sequence below.

Field facts

Date: [record]

Start / finish / breaks: [record]

People and labor hours: [record]

Weather and site conditions: [record]

Starting inventory -> accepted output -> rejected / reworked -> remaining:
[record]

Measurements and configuration used: [record]

The scene

Which adjustment made the next piece easier to repeat?

[Write the actual scene in your own words: what you saw, tried, changed, and understood.]

Evidence and next step

Photo or video IDs, captions, receipts, and notes: [record]

What changed in the plan and why: [record]

Next step: [record]

Day 06 — The tree has a say

Planned focus

Update member assignments and the remaining usable stock count.

This is a proposed focus, not a claim that the work occurred on this day.
Record the actual sequence below.

Field facts

Date: [record]

Start / finish / breaks: [record]

People and labor hours: [record]

Weather and site conditions: [record]

Starting inventory -> accepted output -> rejected / reworked -> remaining:
[record]

Measurements and configuration used: [record]

The scene

Where did taper or a defect force a decision?

[Write the actual scene in your own words: what you saw, tried, changed, and understood.]

Evidence and next step

Photo or video IDs, captions, receipts, and notes: [record]

What changed in the plan and why: [record]

Next step: [record]

Day 07 — Halfway through the clock

Planned focus

Reconcile accepted members, time spent, and the work still ahead.

This is a proposed focus, not a claim that the work occurred on this day.
Record the actual sequence below.

Field facts

Date: [record]

Start / finish / breaks: [record]

People and labor hours: [record]

Weather and site conditions: [record]

Starting inventory -> accepted output -> rejected / reworked -> remaining:
[record]

Measurements and configuration used: [record]

The scene

What remains uncertain at the halfway point?

[Write the actual scene in your own words: what you saw, tried, changed, and understood.]

Evidence and next step

Photo or video IDs, captions, receipts, and notes: [record]

What changed in the plan and why: [record]

Next step: [record]

Day 08 — The first panel family

Planned focus

Check the first accepted A-A-A batch against its reference geometry.

This is a proposed focus, not a claim that the work occurred on this day.

Record the actual sequence below.

Field facts

Date: [record]

Start / finish / breaks: [record]

People and labor hours: [record]

Weather and site conditions: [record]

Starting inventory -> accepted output -> rejected / reworked -> remaining:
[record]

Measurements and configuration used: [record]

The scene

When did loose members begin to feel like a building?

[Write the actual scene in your own words: what you saw, tried, changed, and understood.]

Evidence and next step

Photo or video IDs, captions, receipts, and notes: [record]

What changed in the plan and why: [record]

Next step: [record]

Day 09 — The second panel family

Planned focus

Check B-A-B panels and compare their fixture and joint requirements.

This is a proposed focus, not a claim that the work occurred on this day.
Record the actual sequence below.

Field facts

Date: [record]

Start / finish / breaks: [record]

People and labor hours: [record]

Weather and site conditions: [record]

Starting inventory -> accepted output -> rejected / reworked -> remaining:
[record]

Measurements and configuration used: [record]

The scene

What changed when the short members entered the pattern?

[Write the actual scene in your own words: what you saw, tried, changed, and understood.]

Evidence and next step

Photo or video IDs, captions, receipts, and notes: [record]

What changed in the plan and why: [record]

Next step: [record]

Day 10 — Forty triangles, if ready

Planned focus

Reconcile the actual panel count and photograph labels and spares.

This is a proposed focus, not a claim that the work occurred on this day.

Record the actual sequence below.

Field facts

Date: [record]

Start / finish / breaks: [record]

People and labor hours: [record]

Weather and site conditions: [record]

Starting inventory -> accepted output -> rejected / reworked -> remaining:
[record]

Measurements and configuration used: [record]

The scene

What almost escaped the final panel check?

[Write the actual scene in your own words: what you saw, tried, changed, and understood.]

Evidence and next step

Photo or video IDs, captions, receipts, and notes: [record]

What changed in the plan and why: [record]

Next step: [record]

Day 11 — Ready to assemble

Planned focus

Review the base and readiness against the actual supported assembly plan.

This is a proposed focus, not a claim that the work occurred on this day.

Record the actual sequence below.

Field facts

Date: [record]

Start / finish / breaks: [record]

People and labor hours: [record]

Weather and site conditions: [record]

Starting inventory -> accepted output -> rejected / reworked -> remaining:
[record]

Measurements and configuration used: [record]

The scene

Which preparation mattered more than you expected?

[Write the actual scene in your own words: what you saw, tried, changed, and understood.]

Evidence and next step

Photo or video IDs, captions, receipts, and notes: [record]

What changed in the plan and why: [record]

Next step: [record]

Day 12 — The frame begins to rise

Planned focus

Record the stage, supports, personnel, weather, and measured reference checks.

This is a proposed focus, not a claim that the work occurred on this day. Record the actual sequence below.

Field facts

Date: [record]

Start / finish / breaks: [record]

People and labor hours: [record]

Weather and site conditions: [record]

Starting inventory -> accepted output -> rejected / reworked -> remaining: [record]

Measurements and configuration used: [record]

The scene

What did the first assembled stage reveal?

[Write the actual scene in your own words: what you saw, tried, changed, and understood.]

Evidence and next step

Photo or video IDs, captions, receipts, and notes: [record]

What changed in the plan and why: [record]

Next step: [record]

Day 13 — Approaching closure

Planned focus

Document fit and corrections; proceed only through the actual assembly plan.

This is a proposed focus, not a claim that the work occurred on this day.

Record the actual sequence below.

Field facts

Date: [record]

Start / finish / breaks: [record]

People and labor hours: [record]

Weather and site conditions: [record]

Starting inventory -> accepted output -> rejected / reworked -> remaining:
[record]

Measurements and configuration used: [record]

The scene

Where did the last relationships agree or disagree?

[Write the actual scene in your own words: what you saw, tried, changed, and understood.]

Evidence and next step

Photo or video IDs, captions, receipts, and notes: [record]

What changed in the plan and why: [record]

Next step: [record]

Day 14 — The honest finish line

Planned focus

Record the achieved condition, actual totals, remaining work, and final photographs.

This is a proposed focus, not a claim that the work occurred on this day. Record the actual sequence below.

Field facts

Date: [record]

Start / finish / breaks: [record]

People and labor hours: [record]

Weather and site conditions: [record]

Starting inventory -> accepted output -> rejected / reworked -> remaining: [record]

Measurements and configuration used: [record]

The scene

What can you now claim, and what remains to be done?

[Write the actual scene in your own words: what you saw, tried, changed, and understood.]

Evidence and next step

Photo or video IDs, captions, receipts, and notes: [record]

What changed in the plan and why: [record]

Next step: [record]

25. Count the work the wood actually replaced

The discovery notes use an illustrative store price and a proposed production rate to ask whether wedge-making can have meaningful value. Preserve those figures as the origin of the question, with their original date and uncertainty. Replace them with receipts and measured labor when making a claim about this build.

Separate cash spent, reusable equipment acquired, consumables used, and labor hours. A saw bought for this project still has value afterward, while fuel and oil used during the project do not. A comparison can show either initial cash requirement or allocated project cost, but it should say which one it shows.

Count labor from setup through completed accepted parts. Record maintenance, movement, fixture work, inspection, failed attempts, and cleanup where they belong. A rate measured during one productive afternoon does not establish the average rate for the whole frame. The book can show both the good run and the full ledger.

To compare with purchased lumber, define the product the purchased material would have supplied. Volume equality alone does not establish equal structural function, connection cost, dimensional stability, or finish quality. The strongest comparison is between complete, specified alternatives that serve the same purpose.

Use the same discipline for the “two trees” claim. It describes the proposed source of wedge members. List added seam material, fasteners, supports, and anything bought for the base separately. The extra rows do not weaken the experiment; they show its actual boundary.

The economic ending can remain useful even if it is mixed. The method may save a machine while requiring more handling. It may preserve wood

while demanding a more involved connection. Those tradeoffs tell a future builder where the method fits and where a conventional alternative may fit better.

Scene study - Reading the math labels

The source film separates observations, calculations, planning choices and estimates. The labels and values below are reproduced from that lesson to explain its visual language; they are not new entries in this book's build log.

Read the labels before the totals

- MEASURED fuel while ripping: 2 tanks an hour, one timed sitting
- MEASURED cutting rate: 30 struts in a 6-hour afternoon
- DERIVED ripping time: 120 struts at 5.0 an hour = 24 hours
- DECIDED the plan: felling days 1-2, bucking days 3-4, ripping days 5-8
- ESTIMATED the saw's tank: 0.28 to 0.37 L; the listings disagree
- NOT METERED fuel for felling and bucking: left out, not guessed

What the displayed conclusion includes

So the fuel is a range, and it is the ripping only. This is the scope of the source lesson's calculation. Compare its assumptions with your own recorded tank size, refill volumes, operating hours and cutting rate before reusing a total.

The still below shows the original math presentation. The same values are typeset above so the explanation remains readable on a printed page.

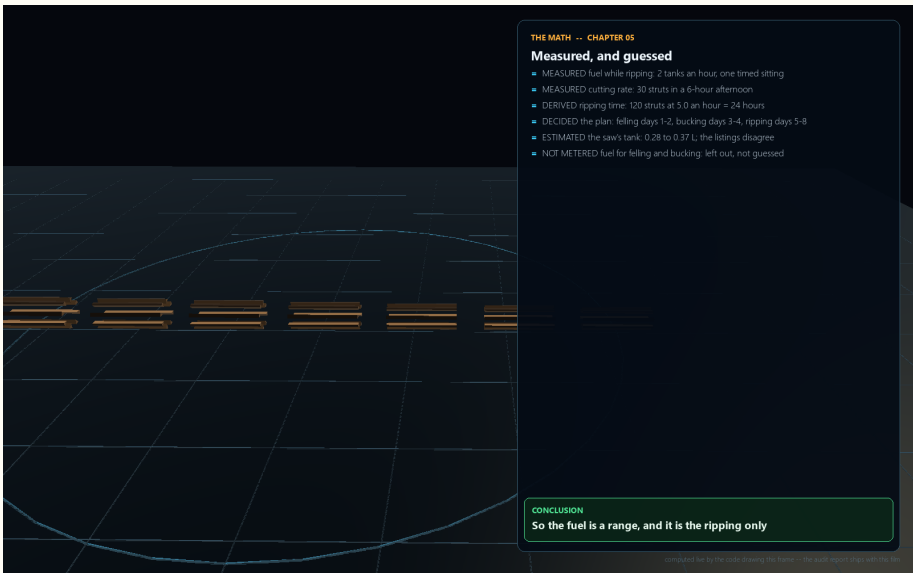


Figure 1

The source lesson's math overlay preserves measured, derived, decided, estimated and unmetered labels. This is a snapshot of its current example; the source values are printed above for reading at book size.

Source: Rendered from Two Trees, All at Once / assume at 95%, using the original video scene and camera (math overlay). Source model assumptions, not verified field results. Timing uses authored chapter durations; source hashes and the complete snapshot recipe are saved with this image.

26. What two trees will have taught me

At the beginning, the idea fits in one sentence: use the wedge that a tree already contains to make the repeated parts of a dome. By the end, that sentence should have acquired measurements, joints, mistakes, photographs, and a result.

I want the last chapter to return to the first inventory. How many sections did the actual trees supply? How many blanks became accepted members? Which allowance changed after the first physical trial? How much of the fortnight went into sawing, how much into making the geometry repeatable, and how much into joining the panels?

Those answers cannot be written in advance. What can be written is the standard they must meet. The count should reconcile. The final model should match the built configuration. The photographs should show the important details. The story should preserve the moments where the wood or the assembly contradicted the expectation.

If the frame comes together within the target, the account should explain what preparation and conditions made that possible. If it takes longer, the account should explain the difference. The discovery has value when another person can understand and evaluate it; a protected deadline would make the explanation less useful.

The closing scene belongs to the author after the work. Return to one object from the opening: a source stump, an unused wedge, the saw, or the first sketch. Describe it as it is now, after the experiment. Let the measured result and the lived scene occupy the same page.

PART / reference / draft

Part VII — The wider project and the bench reference

The dome in this book sits within a larger set of experiments.

The atlas shows how the tools connect. The reference pages keep the method usable when the narrative has been read and the next decision is on the bench.

27. The tools that made the questions visible

DomeSim is a workshop of related tools. Their value to the book is that they make different parts of the same question visible. A film can introduce an idea; a physical solver can inspect a joint; a site designer can show how a frame might become a place to live.

Project tool or study	Role in the book	Suggested image
Raw Wedge Dome	Physical members, orientations, cut planes, seams, and jig	Four orientations and a close seam
2V geometry / Masterclass	Sphere subdivision, factors, panel families	Topology beside panel ownership
Frankendome	Earlier experiments with varied triangular members	Dated prototype photograph and explanatory frame
Hex masterclass	Hexagon/pentagon cages and changes in panel shape	A labeled model plate
Zome masterclass	A related form built from rhombi and generating directions	Polar zome plate
Dome Creator	Panels, interior concepts, services, and site arrangement	Selected saved scene with configuration
Dome Forge	Configurable build variants, groups, layers, and jigs	A chosen design and layer comparison
Assembly Line	Material flow and the work of making repeated units	A scene tied to measured production stages
Presenter / video lessons	Explanatory sequence and narrative evidence	A still that communicates one claim

Project tool or study	Role in the book	Suggested image
Beat and voice tools	Narration experiments and reusable short sequences	Script excerpt with the finished page it informed

The atlas is not a claim that every depicted structure can use the same 120-piece inventory. The 2V panelized hemisphere owns that count. A hex cage and a zome have their own topology and material requirements. Keep their configuration labels visible when they appear in the same spread.

For each selected project still, retain the source tool, configuration, camera, date, and a caption saying what the image demonstrates. Rendered imagery can explain geometry very well. Actual build photographs establish what the author made. Use each where it carries the evidence the page needs.

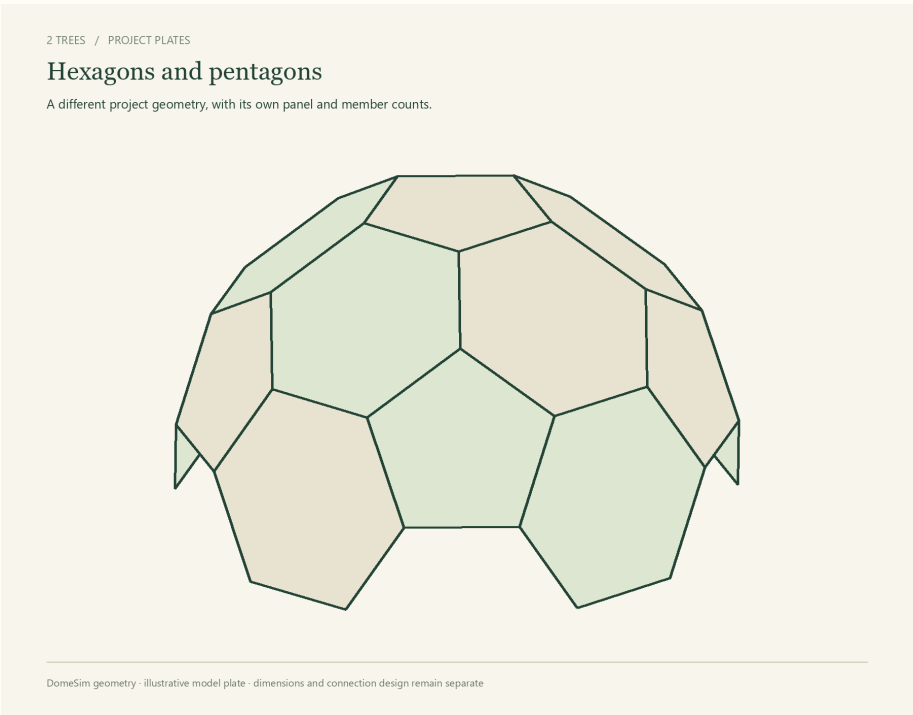


Figure 1

The hex tool's truncated-icosahedron dome selection; a distinct construction study.

Source: Rendered from existing DomeSim model geometry; see `two_trees_codex/render.py`. Print colors are illustrative.

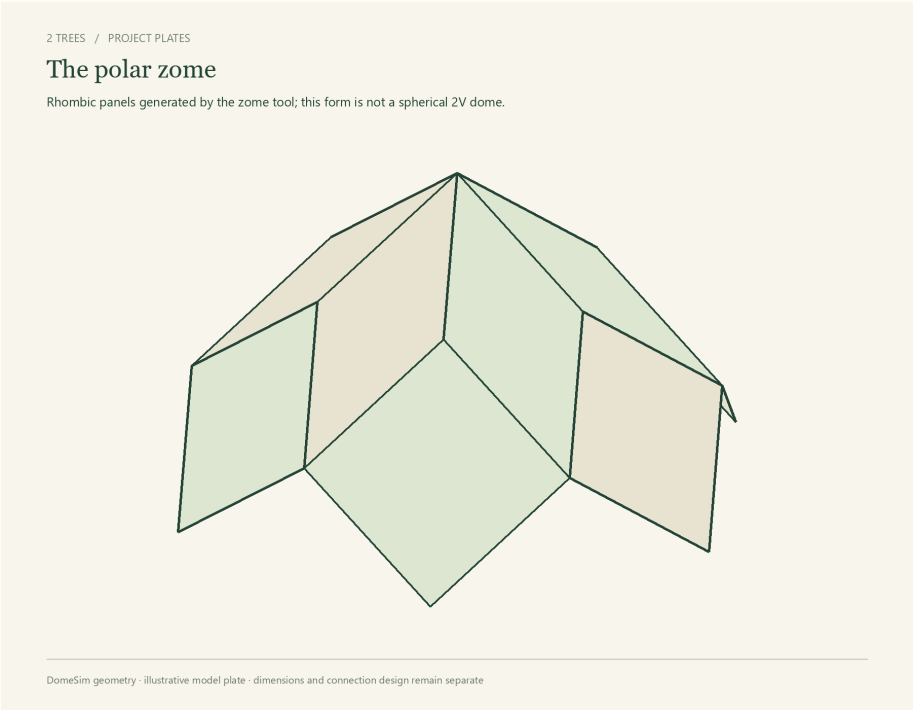


Figure 2

The zome tool's default polar zome; a distinct rhombic-panel geometry.

Source: Rendered from existing DomeSim model geometry; see `two_trees_codex/render.py`. Print colors are illustrative.

Illustration plan — give each image a job

Placement	Figure	Evidence or purpose
Title / Part I	Woodland dome artwork	Conceptual mood, clearly labeled as artwork
Chapter 2	Frankendome before-and-after	Author photographs needed to establish chronology
Chapter 3	First wedge and first corner	Actual discovery record needed
Chapters 5–6	Ideal mesh and independent timber panels	Explain 65 edges versus 120 members
Chapters 7–10	Dimension and inventory plates	Show the two sizing directions and kerf allowance
Chapter 11	True circular sector	Preserve the arc, distinguish diameter from radius
Chapters 12–14	Trunk labels, defects, moisture record	Actual field evidence needed
Chapter 16	Four raw-wedge orientations	Generated from the existing physical solver
Chapters 17–18	Corner, cut planes, fixture stages	Simulator plates plus full-size prototype photographs
Chapters 19–22	Two panel families, base map, supported stages	Model reference beside the actual sequence
Chapter 23	Plan and section of a proposed interior	Concept study with the chosen configuration

Placement	Figure	Evidence or purpose
Fourteen-day journal	One revealing image per day	What changed, including errors and corrections
Chapter 27	Geodesic, wedge, hex cage, polar zome	Comparison of project models, not interchangeable cut lists

Every image needs a caption, provenance, and a reason to occupy the page. For artwork, say that it is conceptual. For a simulator plate, name its configuration. For a photograph, record date and subject. For an explanatory diagram, show units and the definitions used.

Choose full-page plates for geometry that needs space and part openers that need a pause. Use smaller images beside a calculation when the reader must compare numbers and shapes at the same time. A photograph of an unmarked pile should not substitute for a measured inventory diagram.

Worksheet A — Source tree and section register

Date: [record] Tree ID: [record] Species and identification basis: [record]

Section ID	Usable length	Large / small diameter	Defects / sweep	Potential blanks	Accepted blanks	Photo ID
T1-S01						
T1-S02						
T1-S03						
T1-S04						
T1-S05						
T1-S06						
T1-S07						
T1-S08						

Duplicate this page for the second tree. Record end trims and removed defect lengths separately from the retained sections. Photograph every rejected region whose removal materially changes the yield.

Reconciliation: retained section length + separating kerfs + removed material = measured original usable span, within the documented measurement uncertainty.

Worksheet B — Member and panel traceability

Configuration file and date: [record]

Member ID	Source section / sector	Edge family	Required blank length	Measured available length	Orientation	Acceptance / reason
Panel ID	Family	Three member IDs	Fixture version	Reference checks	Acceptance	Intended position

Use the actual solver's member IDs in the cut schedule. Do not relabel pieces solely to make a picture easier to read; create a legend that relates the readable label to the stored ID.

Before assembly, total the accepted panels by family and reconcile their members with the material ledger. Keep the accepted spare inventory visible after the forty panels are complete.

Worksheet C — Time, purchases, and what remains

Date	Work category	People / labor hours	Accepted output	Consumables	Rework / delay	Record or receipt
Purchase	Cash paid	Used this build	Reusable afterward	Allocation method	Receipt date	

Record pre-clock work on the same basis as work during the fortnight. Report calendar elapsed time and labor hours separately. Two people working six hours contribute twelve labor hours during six elapsed hours.

At the final inventory, record installed material, accepted unused stock, unprocessed sections, test pieces, and rejected material. The remaining wood is part of the experiment's result.

Glossary — the words beside the cuts

A / B: This book's long / short nominal geodesic edge families.

Blank: The stock available before a member's final cuts, including necessary process allowance.

Chord factor: The ratio of an ideal edge length to sphere radius.

Decagon: The ten-sided base polygon of this hemisphere.

Edge: A unique line segment in the geometric mesh; not necessarily one wooden member.

Fold: The smaller angle between outward normals of adjacent panels.

Kerf: Material removed across the width of a saw cut. Bucking kerf affects section-length inventory; ripping kerf affects sector recovery.

Member: One finished structural wooden part, with its physical geometry and identity.

Panel: One independently assembled triangular frame, owning three members.

Pinwheel: A cyclic end-to-side relationship within a panel, with each member receiving one neighbor and terminating against another.

Radial sector: A region bounded by two radii and their circular arc; one eighth has a forty-five-degree included angle.

Seam: A shared interface between two panels. This hemisphere has fifty-five interior seams and ten exposed base edges.

Stock envelope: The longitudinal extent needed to contain the physical member before any extra process allowance.

Zone: A related form based on a zonohedral construction; it has its own geometry and is not this spherical 2V frame.

Sources, assumptions, and corrections to resolve

The project files provide the design history and computations for this draft. The author's brief provides the intended two-tree inventory, the two-week target, and the saw identification. These are plans and author statements until field records establish measured results.

Source	What it supports	Publication check
discovering-the-wedge.txt	Frankendome history, discovery narrative, original value and yield hypotheses	Confirm dates, original photos, and which observations were measured
dome-spec-wedges.txt	2V topology, independent panels, sector and seam concepts	Reconcile earlier spec with current physical solver
two_v_demo/geometry.py	Exact chord factors and generated hemisphere	Retain the actual code revision used
geodesic_raw_wedge_dome_dihedral.py	Physical members, cuts, seams, orientations, and fixture settings	Retain the configuration used for fabrication
two_v_demo/raw_wedge_bridge.py	Shared access to the physical simulator and meshes	Used by the book's render and calculation adapters
two_v_demo/wedge_geometry.py	Earlier sector, tree-yield and simplified pinwheel studies	Identify differences from this book's six-foot section plan
two_v_demo/hex_geometry.py and zome_geometry.py	Alternative project forms	Label each form with its own topology

External reference links:

- Husqvarna 120 product page
(<https://www.husqvarna.com/us/products-category/chainsaws/120/>) — manufacturer reference; check the purchased model and its own manual.
- Husqvarna product-number identification
(<https://us-support.husqvarna.com/en/petrol-chainsaw/KA-70145>) — how to identify the specific variant.
- USDA Wood Handbook
(<https://research.fs.usda.gov/fpl/wood-handbook>) — material properties, moisture, grading, fastening, and design background.
- USFS saw training guide (<https://www.fs.usda.gov/t-d/pubs/htmlpubs/hmo6672805/page01.htm>) — training context for field saw operations.

The printed factors and worked results are rounded for reading. Live calculations use generated geometry. The nominal six-foot edge example is not a final cut prescription. The eighty-strut-per-tree example in an earlier timber demonstration uses a different source length and sectioning method from the sixty-four-blank plan here.

The source notes' trunk diameter text includes an apparent feet/inches typo. This draft uses the explicitly named eight-inch and twelve-inch diameter examples as declared mathematical examples, not as measured dimensions of the author's trees. No unverified claim of invention priority, universal recovery percentage, saw capability, or structural rating is established by this manuscript.

Before publication, replace open field prompts, verify the saw identification from the machine, date all prices, reconcile the final counts, update the physical configuration, and have the construction details reviewed for their intended use. Preserve the corrections in the explanation where they changed the method.